

Algorithm Cheat Sheet — All Templates + Java Syntax

Conventions, Skeleton & Core API

Generic Test Harness

```
class Question {
    // --- PROBLEM-SOLVING PROCESS ---
    // 1. INPUT / OUTPUT : types, sizes, ranges; what to return; examples.
    // 2. HIGH-LEVEL ALGO : pick the pattern (two-ptr / BFS / DP / ...); state the idea in 1 line.
    // 3. IMPLEMENTATION : data structures, loop/recursion, invariants.
    // 4. EDGE CASES : empty, size 0/1, duplicates, negatives, overflow, all-same, not-found
    // 5. COMPLEXITY : time & space; is it within limits?
    // 6. OPTIMIZATION : reduce time/space (hashing, two-pointer, rolling array, pruning, early exit).
    private int data;
    public Question(int data) {
        this.data = data;
    }
    public void solve() {
        // TODO
    }
}

public class Solution {
    public static void main(String[] args) {
        Question q = new Question(0);
        q.solve();
    }
}
```

Naming Conventions (enforced across all template files)
Common Java API Cheat-Sheet

```
Map<Integer, Integer> map = new HashMap<>(); // key - value
Map<Integer, List<Integer>> adj = new HashMap<>(); // adjacency / buckets
Set<Integer> set = new HashSet<>(); // membership
List<Integer> list = new ArrayList<>(); // dynamic array
Deque<Integer> stack = new ArrayDeque<>(); // STACK: push / pop / peek
Deque<Integer> queue = new ArrayDeque<>(); // QUEUE: offer / poll / peek / ArrayDeque beats

java.util.Stack / LinkedList
PriorityQueue<Integer> minHeap = new PriorityQueue<>(); // min-heap (default)
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<int> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // by field, min-first
TreeMap<Integer, Integer> tmap = new TreeMap<>(); // sorted keys: floor/ceiling/first/last
TreeSet<Integer> tset = new TreeSet<>(); // sorted set: floor/ceiling/higher/lower
LinkedHashSet<Integer> lset = new LinkedHashSet<>(); // insertion-ordered set (LIFO buckets)
Random rand = new Random(); // defaults to 0..n-1 (RandomizedSet)

int[] arr = new int[n]; // defaults to 0
boolean[] seen = new boolean[n]; // defaults to false
int[][] grid = new int[m][n]; // 2D, all 0
int[] dir = {1,-1,0,0}; dc = {0,0,1,-1}; // DOWN,UP,RIGHT,LEFT; iterate for (dir=0..3) { int d=dir[j]; <dc[d]>

// --- Map
map.put(k, v);
map.getDefault(k, 0); // safe read with default
map.merge(k, 1, Integer::sum); // + counting frequency (most common)
map.computeIfAbsent(k, x -> new ArrayList<>()).add(v); // + build adjacency / buckets
map.containsKey(k, v); // write only first occurrence (keep earliest index)
map.containsKey(k); map.remove(k);
for (Map.Entry<Integer, Integer> e : map.entrySet()) { e.getKey(); e.getValue(); e.setValue(v); }
map.keySet(); map.values();
// --- Set
set.add(x); set.contains(x); set.remove(x);
// --- List
list.add(x); list.get(i); list.set(i, x); // + pop in backtracking
list.remove(list.size() - 1); // defaults to first
list.size(); list.isEmpty(); list.addAll(others);
Collections.sort(list); // ascending
list.sort((a, b) -> b - a); // custom / descending
Collections.reverse(list); // reverse path after building it backwards
```

```
// Deque as STACK (LIFO) // Deque as QUEUE (FIFO)
stack.push(x); queue.offer(x);
int top = stack.pop(); queue.offer(x);
int look = stack.peek(); int look = queue.peek();
stack.isEmpty(); queue.isEmpty();
// Deque BOTH ENDS - needed for the MONOTONE DEQUE (sliding-window max/min)
// and for add-first-style (zigzag BFS)
deque.offerLast(x); deque.pollLast(); // back (push/pop/peek tail)
deque.offerFirst(x); deque.pollFirst(); // front (push/pop/peek head)
deque.addLast(x); deque.addFirst(x); // same as offer*, throw if capacity-bound
// PriorityQueue (heap)
pq.offer(x); int best = pq.poll(); int look = pq.peek(); pq.size();
pq.remove(x); // - O(n) removal of an arbitrary element (lazy-deletion alternative in #480)
```

```
Integer.MAX_VALUE; // 2147483647 (2^31 - 1) - sentinel for "min so far"
Integer.MIN_VALUE; // -2147483648 (-2^31) - sentinel for "max so far"
Long.MAX_VALUE; // when sums can overflow int, accumulate in long
int k = 1 + (j - 1) / 2; // avoid (1 + j) / 2 overflow in binary search
Integer.compare(a, b); // - overflow-safe comparator; use instead of (a < b)
(a, b) -> Integer.compare(a[1], b[1]); // - safe PriorityQueue/Arrays.sort comparator
Double.compare(a, b); // - comparator for double[] heaps (probability, median)
```

```
ch = 'a'; // 'a'..'z' -> 0..25 (lowercase bucket index)
ch = '0'; // '0'..'9' -> 0..9 (digit value)
(char) ('a' + i); // 0..25 -> 'a'..'z'
num = num * 10 + (ch - '0'); // - build a number digit by digit
Integer.parseInt("123"); // String -> int (throws on bad input)
Integer.valueOf(123); // int -> Integer (boxed)
String.valueOf(123); // int/char/etc -> String
Character.getNumericValue('7'); // char classification
Character.isDigit(ch); Character.isLetter(ch); Character.isLetterOrDigit(ch);
Character.isWhitespace(ch); Character.toLowerCase(ch); Character.toUpperCase(ch);
Character.toUpperCase(ch);
```

```
s.length(); s.charAt(i); s.substring(i, j); // [i, i)
s.toCharArray(); s.split(""); s.equals(t); s.compareTo(t);
s.indexOf("ab"); s.isEmpty();
s.startsWith("ab"); s.endsWith("z"); s.contains("x"); // prefix / suffix / substring tests
s.toLowerCase(); s.toUpperCase(); // case-normalize (e.g. case-insensitive compare)
s.trim(); s.replace('a', 'b'); // strip ends / swap chars / repeat
StringBuilder builder = new StringBuilder();
builder.append(x); builder.reverse(); builder.toString();
builder.charAt(i); builder.setCharAt(i, c); builder.deleteCharAt(i);
builder.length();
String.join(",", list); // list = "a,b,c"
```

```
Arrays.sort(arr); // ascending in place
Arrays.sort(intervals, (a, b) -> a[0] - b[0]); // 2D by first field
Arrays.fill(dp, Integer.MAX_VALUE); // send a dp/dist array
Arrays.equals(window, need); // - element-wise array compare (anagram check)
Arrays.copyOf(arr, len); Arrays.copyOfRange(arr, i, j);
System.arraycopy(src, srcPos, dst, dstPos, len); // - fast block copy (merge step)
Arrays.asList(1, 2, 3);
int sum = Arrays.stream(arr).sum(); // fixed-size List view
int max = Arrays.stream(arr).max().getAsInt();
// List-Integer -> int[]
int[] a = List.stream().mapToInt(Integer::intValue).toArray();
List<Integer> l = Arrays.stream(a).boxed().collect(Collectors.toList());
```

```
Math.max(a, b); Math.min(a, b); Math.abs(x);
Math.pow(a, b); Math.sort(x); Math.ceil(x); Math.floor(x);
n & (n - 1); // clear lowest set bit
n & (-n); // isolate lowest set bit
1 << i; // bit i set (use 1 << i for i > 31)
(n >> i) & 1; // read bit i
(n & 1) == 0; // - even (last bit 0)
(n & 1) == 1; // - odd (last bit 1)
n >> 1; // - divide by 2 (drop last bit); n << 1 = multiply by 2
Integer.bitCount(n); // # of set bits
Integer.toBinaryString(n);
```

```
tmap.firstKey(); tmap.lastKey();
tmap.floorKey(x); // largest key <= x tmap.ceilingKey(x); // smallest key >= x
tmap.lowerKey(x); // largest key < x tmap.higherKey(x); // smallest key > x
tset.floorKey(x); tset.ceiling(x); tset.lower(x); tset.higher(x); tset.first(); tset.last();
```

```
// --- NEGATIVE-SAFE MODULO --- replaces the manual ((x % n) + n) % n
Math.floorMod(x, n); // always in [0, n-1] even when x < 0
(prefix-sum % k)
// int[] INSTEAD Of HashMap for a bounded key space -
int[] count = new int[26]; // O(1) no-boxing freq vs
HashMap<Character, Integer> count[ch = 'a']; // also faster to compare: Arrays.equals(a, b)
// --- 1D ROLLING ARRAY for DP --- drop a dimension when row i only needs row i-1
int[] dp = new int[n + 1]; // O(n) space instead of O(m*n)
for (...) for (int j = target; j >= w; j--) dp[j] += dp[j - w]; // O/1 knapsack: iterate j DOWN
// --- FIXED-SIZE HEAP for top-k --- O(n log k) beats sorting O(n log n)
if (heap.size() > k) heap.poll(); // keep only k best; min-heap for "k largest"
// --- LAZY DELETION over pq.remove(obj) --- pq.remove is O(n); skip stale entries instead
if (entry[0] != dist[node]) continue; // Dijkstra: ignore outdated heap entries
// --- ArrayDeque over Stack / LinkedList --- faster, no synchronization, both ends O(1)
Deque<Integer> stack = new ArrayDeque<>(); // never use java.util.Stack
// --- StringBuilder over String += in a loop --- O(n) vs O(n^2)
StringBuilder builder = new StringBuilder(); // append, then builder.toString() once
// --- EARLY EXIT / PRUNING --- return the moment the answer is decided
if (found) return result; // e.g. backtracking bounds
```

Template Index

Two Pointers

Pattern 1 — Opposite Two Pointers

```
Variables:
i = left pointer · j = right pointer · scan range [i, j] closed
Pseudocode:
set i to the left end, j to the right end
while the pointers haven't crossed:
    if the current pair hits the target:
        record it, then step both ends inward
    else if the value is too small:
        move the left end right to grow it
    else (too large):
        move the right end left to shrink it
```

```
int i = 0, j = n - 1;
while (i < j) {
    if (condition == target) {
        // record */ i++; j--;
    } else if (tooSmall) {
        i++;
    } else {
        j--;
    }
}
```

Sliding Window

The Three Templates

```
Variables:
i = left edge (inclusive) · j = right edge / loop variable (inclusive), window = [i, j], size = j - i + 1 · n = array length · k = target window size · result = answer (max length, or min length seeded to MAX)
Pseudocode:
```

```
FIXED window size of k:
i = 0
for j = 0..n-1:
    add nums[j] to window state
    if window size (j - i + 1) == k:
        record the window (it is exactly size k)
        remove nums[i] from state; i = i + 1
```

```
MAX variable window (longest valid):
i = 0; result = 0
for j = 0..n-1:
    add nums[j] to state
    while window violates the constraint:
        remove nums[i] from state; i = i + 1
    result = max(result, j - i + 1) (window is valid here)
```

```
MIN variable window (shortest valid):
i = 0; result = +infinity
for j = 0..n-1:
    add nums[j] to state
    while window is valid:
        result = min(result, j - i + 1) (record while still valid)
        remove nums[i] from state; i = i + 1 (then shrink to try smaller)
```

```
int i = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    if (i - 1 + 1 == k) {
        // 2. record (window is exactly size k)
        // 3. remove nums[i] from state
        i++;
    }
    int i = 0, result = 0;
    for (int j = 0; j < n; j++) {
        // 1. add nums[j] to state
        while (/* window exceeds constraint */) {
            i++;
        }
        result = Math.max(result, j - i + 1);
    }
    int i = 0, result = Integer.MAX_VALUE;
    for (int j = 0; j < n; j++) {
        // 1. add nums[j] to state
        while (/* window exceeds constraint */) {
            result = Math.min(result, j - i + 1);
            i++;
        }
    }
}
```

Frequency Map Trick (used by 567 and 76)

```
Variables:
need[c] = how many more of char c the window still needs (positive = still short, zero/negative = surplus) · required = total chars still missing, hits 0 when the window covers all of it · i = left edge, j = right edge
Pseudocode:
```

```
Expanding (add char at j):
if need[char] was still positive (> 0): required = required - 1 (filled one more)
need[char] = need[char] - 1
Shrinking (remove char at i):
if need[char] was already 0 or surplus (>= 0): required = required + 1 (now short one)
need[char] = need[char] + 1
i = i + 1
if (need[s.charAt(j)] == -> 0) required--;
if (need[s.charAt(i)] += >= 0) required++;
i++;
```

Prefix Sum + HashMap

Core Template

```
Variables:
map = prefixSum-count or index seen so far · sum = running prefix sum · lookup = the complement prefix we search for · i = current scan position
Pseudocode:
```

```
create empty map of prefix -> (count or index)
seed map with prefix 0 (count=1 for counting, index=-1 for length)
sum = 0
for each index i:
    add nums[i] to sum
    lookup = transform(sum) // sum-k, sum/k, etc.
    use map.get(lookup) to update result
    store sum (or sum/k) into map with count or index i
```

```
Map<Long, Integer> map = new HashMap<>();
map.put(0L, ???);
long sum = 0;
for (int i = 0; i < nums.length; i++) {
    sum += nums[i];
    long lookup = transform(sum);
    map.put(storeKey(sum), storeValue(i)); // what to store (sum or sum/k - count or index)
}
```

Binary Search

The Only Template — [i, j], find first k with condition(k)

```
Variables:
[i, j] = half-open search range (answer lives in here) · k = midpoint · condition(k) = monotone test, false...false then TRUE...true; answer is the first k it holds
```

```
Pseudocode:
i = 0, j = length # half-open [i, j)
while range non-empty (i < j):
    k = midpoint of i and j
    if condition(k) is false: answer is to the right, set i = k + 1
    else: k might be the answer, set j = k
```

```
int i = 0, j = nums.length;
while (i < j) {
    int k = i + (j - i) / 2;
    if (!condition(k)) {
        i = k + 1;
    } else {
        j = k;
    }
}
```

How to Choose

Sorting

Template 1 — Merge Sort

```
Variables:
[start, end] half-open range being sorted · mid = split point · temp = scratch buffer · i = left scan pointer [start, mid] · j = right scan pointer [mid, end] · k = write pointer into temp
Pseudocode:
```

```
mergeSort(nums, start, end):
    if range has 0 or 1 elements: return // already sorted
    mid = midpoint of [start, end]
    mergeSort left half [start, mid]
    mergeSort right half [mid, end]
    merge the two sorted halves

merge:
    i = start, j = mid, k = start
    while either half has elements left:
        pick smaller front of the two halves (left wins ties - stable)
        write it to temp[k], advance that half's pointer and k
        copy temp back into nums[start, end]

public void mergeSort(int[] nums) {
    mergeSort(nums, 0, nums.length, new int[nums.length]);
}

private void mergeSort(int[] nums, int start, int end, int[] temp) {
    if (end - start <= 1) return;
    int mid = start + (end - start) / 2;
    mergeSort(nums, start, mid, temp);
    mergeSort(nums, mid, end, temp);
    merge(nums, start, mid, end, temp);
}

private void merge(int[] nums, int start, int mid, int end, int[] temp) {
    int i = start, j = mid, k = start;
    while (i < mid || j < end) {
        if (j == end || (i < mid && nums[i] <= nums[j])) {
            temp[k++] = nums[i++];
        } else {
            temp[k++] = nums[j++];
        }
    }
    for (i = start; i < end; i++) nums[i] = temp[i];
}
```

Template 2 — Quick Sort (3-way Dutch Flag Partition)

```
Variables:
[start, end] half-open range · pivot = chosen partition value · i = boundary of <pivot region · k = scan/cursor pointer · j = boundary of >pivot region (from the right)
Pseudocode:
```

```
quickSort(nums, start, end):
    if range has 0 or 1 elements: return
    pivot = middle element's value
    i = start, k = start, j = end-1 // [start,i)<p | [i,k]==p | [k,j] unknown | (j,end]>=p
    while (k <= j) {
        if nums[k] < pivot: swap into < region, advance k and i
        elif nums[k] == pivot: advance k
        else: swap to > region, shrink j (don't advance k)
        recurse on < region [start, i]
        recurse on > region [k, end] // ==pivot middle is final

public void quickSort(int[] nums) {
    quickSort(nums, 0, nums.length);
}

private void quickSort(int[] nums, int start, int end) {
    if (end - start <= 1) return;
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    quickSort(nums, start, i);
    quickSort(nums, k, end);
}

private void swap(int[] nums, int i, int j) {
    int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
}
```

```
private void quickSort(int[] nums, int start, int end) {
    if (end - start <= 1) return;
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    quickSort(nums, start, i);
    quickSort(nums, k, end);
}

private void swap(int[] nums, int i, int j) {
    int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
}
```

Template 3 — Quick Select (partial sort — k-th element)

```
Variables:
[start, end] half-open range · target = 0-indexed position we want settled · pivot = partition value · i = boundary of <pivot region · k = scan/cursor pointer · j = boundary of >pivot region
Pseudocode:
```

```
quickSelect(nums, start, end, target):
    pivot = middle element's value
    partition into [start,i)<p | [i,k]==p | [k,end]>p // same 3-way as quick sort
    if target < i:
        recurse into left [start, i)
    elif target < k:
        target sits in ==pivot region - return pivot
    else:
        recurse into right [k, end)
```

```
private int quickSelect(int[] nums, int start, int end, int target) {
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    if (target < i) {
        return quickSelect(nums, start, i, target);
    } else if (target <= k) {
        return pivot;
    } else {
        return quickSelect(nums, k, end, target);
    }
}
```

Template 4 — Counting Sort

Variables:
min/max = value range bounds · buckets[v-min] = count of value v · i = write pointer back into arr · v = current value index in buckets

Pseudocode:

```
if array empty: return
find min and max values
buckets = array sized (max-min+1), all zero
for each x in arr: increment buckets[x-min] // tally occurrences
i = 0
for v from 0 to last bucket:
    while bucket v still has count: write (v+min) into arr[i], advance i
```

```
public void countingSort(int[] arr) {
    if (arr.length == 0) return;
    int min = Arrays.stream(arr).min().getAsInt();
    int max = Arrays.stream(arr).max().getAsInt();
    int[] buckets = new int[max - min + 1];
    for (int x : arr) buckets[x - min]++;
    int i = 0;
    for (int v = 0; v < buckets.length; v++)
        while (buckets[v]-- > 0) arr[i++] = v + min;
}
```

Linked List

When to Use — Signal → Pattern

All Four Templates

Variables:
sentinel = dummy node before head · previous = last kept node · current = node under inspection · next = saved successor · slow/fast = 1x/2x walkers · a/b = the two input lists

Pseudocode:

```
make sentinel pointing at head; previous = sentinel, current = head
while current exists:
    if current should be deleted: rewire previous.next to skip current
    else: advance previous to current
    advance current
return sentinel.next
previous = null, current = head
while current exists:
    save next = current.next
    point current backward at previous
    advance previous to current, current to next
return previous as new head
slow = head, fast = head
while fast and fast.next exists:
    advance slow one step, fast two steps
make sentinel; current = sentinel
while both a and b exist:
    splice the smaller head onto current; advance that list
advance current
attach whichever list remains
return sentinel.next
```

```
// 1. DELETE / FILTER - sentinel guards head removal; previous tracks last kept node
ListNode sentinel = new ListNode(0);
sentinel.next = head;
ListNode previous = sentinel, current = head;
while (current != null) {
    if (shouldDelete(current)) {
        previous.next = current.next;
    } else {
        previous = current;
    }
    current = current.next;
}
return sentinel.next;
// 2. REVERSAL - re-wire one node at a time
ListNode previous = null, current = head;
while (current != null) {
    ListNode next = current.next;
    current.next = previous;
    previous = current;
    current = next;
}
return previous;
// 3. FAST / SLOW - two pointers at different speeds
ListNode slow = head, fast = head;
while (fast != null && fast.next != null) {
    slow = slow.next;
    fast = fast.next.next;
}
// 4. MERGE - sentinel collects nodes from two lists in order
ListNode sentinel = new ListNode(0);
ListNode current = sentinel;
while (a != null && b != null) {
    if (a.val <= b.val) {
        current.next = a;
        a = a.next;
    } else {
        current.next = b;
        b = b.next;
    }
    current = current.next;
}
current.next = (a != null) ? a : b;
return sentinel.next;
```

String

Core Idioms

Variables:
count = frequency vector (index = char/digit bucket) · ch - 'a' = 0-25 bucket index · ch - '0' = 0-9 digit value · num = number built so far · builder = encoded/result string · buckets = lists indexed by frequency · expand(i,j) = palindrome length from a center

Pseudocode:

```
idom 1 - char count for letters:
    make count of size 26
    for each char: count[ch-'a'] += 1
idom 2 - char count for digits:
    make count of size 10
    for each char: count[ch-'0'] += 1
idom 3 - char to integer:
    num = 0
    for each char left to right: num = num*10 + (char - '0')
idom 4 - anagram hash key (frequency-based):
    count the 26 letters of s
    build a string from each letter label followed by its count
    return it (same key for all anagrams)
    alternative: sort the chars and use the sorted string as key
idom 5 - bucket sort by frequency:
    count all ASCII chars
    make buckets indexed by frequency
    for each char with count>0: add it to buckets[its count]
    walk buckets from highest frequency down, append each char that many times
idom 6 - expand from center · bounds and chars at i and j match: move i left, j right
    return j - i - 1 (length of palindrome found)
```

```
// 1. CHAR COUNT - lowercase letters
int[] count = new int[26];
for (char ch : s.toCharArray()) count[ch - 'a']++;
// 2. CHAR COUNT - digits 0-9
int[] count = new int[10];
for (char ch : s.toCharArray()) count[ch - '0']++;
// 3. CHAR TO INTEGER - build number digit by digit
int num = 0;
for (int i = 0; i < s.length(); i++)
    num = num * 10 + (s.charAt(i) - '0');
// 4. ANAGRAM HASH KEY - frequency-based (bucket sort style) O(k) vs sort O(k log k)
String hashKey(String s) {
    int[] count = new int[26];
    for (char ch : s.toCharArray()) count[ch - 'a']++;
    StringBuilder builder = new StringBuilder();
    for (int i = 0; i < 26; i++) {
        builder.append((char)('a' + i));
        builder.append(count[i]);
    }
    return builder.toString();
}
chars[] chars = s.toCharArray();
Arrays.sort(chars);
String key = new String(chars);
// 5. BUCKET SORT - sort by frequency without comparison sort
int[] count = new int[128];
for (char ch : s.toCharArray()) count[ch]++;
List<Character>[] buckets = new List[s.length() + 1]; // index = frequency
for (int i = 0; i < 128; i++) {
    if (count[i] > 0) {
        int freq = count[i];
        if (buckets[freq] == null) buckets[freq] = new ArrayList<>();
        buckets[freq].add((char) i);
    }
}
StringBuilder builder = new StringBuilder();
for (int freq = buckets.length - 1; freq > 0; freq--) {
    if (buckets[freq] == null) continue;
    for (char ch : buckets[freq])
        for (int i = 0; i < freq; i++) builder.append(ch);
}
// 6. EXPAND FROM CENTER - palindrome check in O(1) space
int expand(String s, int i, int j) {
    while (i >= 0 && j < s.length() && s.charAt(i) == s.charAt(j)) { i--; j++; }
    return j - i - 1;
}
```

Stack / Queue / Heap

All Templates

Variables:
stack = LIFO ArrayDeque (holds indices for monotone variants) · queue = monotone deque (front = window answer) · minPQ/maxPQ/pq = priority queue (heap) · maxHeap = lower half, minHeap = upper half for median

Pseudocode:

```
STACK: push/pop/peek all at the front (LIFO)
MONO-DECREASING (next greater): while current > stack top, pop and record current as its answer; push index
MONO-INCREASING (next smaller / span): while current < stack top, pop and compute width to new top boundary; push index
MONO-DEQUE (window max): drop back while smaller than current; push; drop front if out of window; front = window max
PRIORITY QUEUE: offer/pol/peek the min (or max with reverse comparator)
TWO HEAPS: push to maxHeap then shift its top to minHeap; rebalance so maxHeap >= minHeap; median from the tops
```

```
// 1. STACK (LIFO) - use ArrayDeque, not Stack class
Deque<Integer> stack = new ArrayDeque<>();
stack.push(x);
stack.pop();
stack.peek();
// 2A. MONOTONE DECREASING STACK - next GREATER element
Deque<Integer> stack = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] < nums[i])
        result[stack.pop()] = nums[i];
    stack.push(i);
}
// 2B. MONOTONE INCREASING STACK - next SMALLER element / span width
Deque<Integer> stack = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] > nums[i]) {
        int height = nums[stack.pop()];
        int width = stack.isEmpty() ? i : i - stack.peek() - 1;
    }
    stack.push(i);
}
// 3. MONOTONE DEQUE - sliding window max/min
Deque<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!queue.isEmpty()) && nums[queue.peekLast()] <= nums[i]
        queue.pollLast();
    queue.offerLast(i);
    if (queue.peekFirst() < i - k + 1) queue.pollFirst();
    if (i >= k - 1) result[i - k + 1] = nums[queue.peekFirst()];
}
// 4. PRIORITY QUEUE
PriorityQueue<Integer> minPQ = new PriorityQueue<>();
PriorityQueue<Integer> maxPQ = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<int> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // custom: sort by index 1
pq.offer(x);
pq.poll();
pq.peek();
// 5. TWO HEAPS - maintain median in O(log n)
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder());
PriorityQueue<Integer> minHeap = new PriorityQueue<>();
void addNum(int num) {
    maxHeap.offer(num);
    minHeap.offer(maxHeap.poll());
    if (maxHeap.size() < minHeap.size())
        maxHeap.offer(minHeap.poll());
}
double findMedian() {
    return maxHeap.size() > minHeap.size()
        ? maxHeap.peek()
        : (maxHeap.peek() + minHeap.peek()) / 2.0;
}
```

BFS (Tree)

The Template — Level-Aware BFS

Variables:
queue = nodes waiting to be processed · size = level snapshot (count of nodes in the current level) · level = current depth counter · node = node polled this step · i = index within the level

Pseudocode:

```
make an empty queue
put root in the queue
level = 0
while the queue is not empty:
    size = how many nodes are in the queue right now (one whole level)
    level = level + 1
    repeat size times (index i = 0..size-1):
        node = take the front of the queue
        process node (level known; i==0 -> first, i==size-1 -> last)
        if node has a left child, add it to the queue
        if node has a right child, add it to the queue

Queue<TreeNode> queue = new ArrayDeque<>();
queue.offer(root);
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();
    level++;
    for (int i = 0; i < size; i++) {
        TreeNode node = queue.poll();
        if (node.left != null) queue.offer(node.left);
        if (node.right != null) queue.offer(node.right);
    }
}
```

When to Use — Signal → Pattern

The One Rule — Snapshot Level Size First

```
Always snapshot the level size BEFORE the inner for loop:
int size = queue.size();
for (int i = 0; i < size; i++) { ... }
```

Without the snapshot, newly added children mix with the current level and i == size-1 / i == 0 checks become meaningless.

DFS / Backtracking

When to Use — Signal → Pattern

All Templates

Variables:
node = current tree node · accumulated/current = state carried down a path · answer/global = best cross-node result · grid[r][c] = cell at row r, col c · state[node] = 0 unseen / in-stack / 2 done · start = first eligible index · current = in-progress candidate · nums[i] = element being chosen

Pseudocode:

```
if node null: return BASE; left=dfs(left); right=dfs(right); return
combine(left,right,node.val)
if node null: return; accumulated+=node.val; if leaf: record; dfs(left,accumulated);
dfs(right,accumulated)
if node null: return 0; left=max(0,dfs(left)); right=max(0,dfs(right));
answer=max(answer, left+right+node.val); return max(left,right)+node.val
if out of bounds or TARGET: return; mark VISITED; recurse into neighbors
if state==1: cycle; if state==2: safe; mark in-stack; recurse neighbors (cycle? return); mark done
if base: record copy; for i from start: skip dups; choose nums[i]; recurse(next);
undo
```

```
// 1. PROCESS CHILDREN FIRST, COMBINE AT NODE (post-order)
private int dfs(TreeNode node) {
    if (node == null) return BASE;
    int left = dfs(node.left);
    int right = dfs(node.right);
    return ...;
}
// 2. PROCESS NODE FIRST, PASS STATE DOWN (pre-order)
private void dfs(TreeNode node, int accumulated) {
    if (node == null) return;
    accumulated += node.val;
    if (isLeaf(node)) { /* record */ return; }
    dfs(node.left, accumulated);
    dfs(node.right, accumulated);
}
// 3. TREE - GLOBAL ANSWER (return up the best single arm; update global across both arms)
int answer = Integer.MIN_VALUE;
private int dfs(TreeNode node) {
    if (node == null) return 0;
    int left = Math.max(0, dfs(node.left));
    int right = Math.max(0, dfs(node.right));
    answer = Math.max(answer, left + right + node.val);
    return Math.max(left, right) + node.val;
}
// 4. GRID DFS - mark visited by mutating grid; 4-directional flood fill
private void dfs(int[][] grid, int r, int c) {
    if (r < 0 || r >= grid.length || c < 0 || c >= grid[0].length) return;
    if (grid[r][c] != TARGET) return;
    grid[r][c] = VISITED;
    dfs(grid, r+1, c); dfs(grid, r-1, c);
    dfs(grid, r, c+1); dfs(grid, r, c-1);
}
// 5. GRAPH DFS - 3-color visited: 0=unseen 1=in-stack 2=done
int[] state;
private boolean dfs(int node) {
    if (state[node] == 1) return true;
    if (state[node] == 2) return false;
    state[node] = 1;
    for (int neighbor : graph.get(node))
        if (dfs(neighbor)) return true;
    state[node] = 2;
    return false;
}
// 6. BACKTRACKING - choose / recurse / undo
private void backtrack(int start, List<Integer> current) {
    if (baseCase) { result.add(new ArrayList<>(current)); return; }
    for (int i = start; i < n; i++) {
        if (skipCondition(i)) continue;
        current.add(nums[i]);
        backtrack(next, current);
        current.remove(current.size()-1);
    }
}
```

Graph

Complexity Legend

BFS / DFS	O(V + E)	visit each vertex and edge once
Topo sort (Kahn)	O(V + E)	each node enqueued once, each edge relaxed once
Union-Find	-O(nα(n)) ~ O(n)	a = inverse Ackermann, effectively constant
Dijkstra	O((V + E) log V)	non-negative weights ONLY

Graph Representation

```
List<List<Integer>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges) {
    graph.get(edge[0]).add(edge[1]);
    graph.get(edge[1]).add(edge[0]);
}
List<List<int>>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges)
    graph.get(edge[0]).add(new int[]{edge[1], edge[2]});
int[] dr = {1, -1, 0, 0};
int[] dc = {0, 0, 1, -1};
```

Core Templates

Variables:
queue = BFS frontier · visited[] = seen flags · level = current distance ring · inDegree[] = remaining prerequisites per node · order = topo result · parent[]/rank[] = union-find forest · dist[] = shortest distance · pq = min-heap on cost · color[] = 2-coloring (<1/0/1)

Pseudocode:

BFS: mark start visited, enqueue; while queue, snapshot level size, pop each, push unvisited neighbors, increment level
TOPO: count inDegree per edge, enqueue all zero-inDegree, pop into order and decrement neighbors, enqueue new zeros
UNION-FIND: init parent[i]=i; find follows parent to root with path compression; union links roots by rank, false if same root
DIJKSTRA: dist[src]=0, push src; pop closest, skip if stale, relax each neighbor and push improved
BIPARTITE: color each component start 0, BFS painting neighbors opposite, return false on same-color clash

```
// 1. BFS - shortest path / level order (track distance by level-size snapshot)
Queue<Integer> queue = new ArrayDeque<>();
boolean[] visited = new boolean[n];
queue.offer(start);
visited[start] = true;
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();
    for (int i = 0; i < size; i++) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (!visited[neighbor]) {
                visited[neighbor] = true;
                queue.offer(neighbor);
            }
        }
    }
    level++;
}

// 2. TOPOLOGICAL SORT - Kahn's BFS
int[] inDegree = new int[n];
for (int[] edge : edges) inDegree[edge[1]]++;
Queue<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) if (inDegree[i] == 0) queue.offer(i);
List<Integer> order = new ArrayList<>();
while (!queue.isEmpty()) {
    int node = queue.poll();
    order.add(node);
    for (int neighbor : graph.get(node))
        if (--inDegree[neighbor] == 0) queue.offer(neighbor);
}

// 3. UNION FIND
int[] parent, rank;
void init(int n) {
    parent = new int[n];
    for (int i = 0; i < n; i++) parent[i] = i;
}

int find(int x) {
    if (parent[x] != x) parent[x] = find(parent[x]);
    return parent[x];
}

boolean union(int x, int y) {
    int px = find(x), py = find(y);
    if (px == py) return false;
    if (rank[px] < rank[py]) { int t = px; px = py; py = t; }
    parent[py] = px;
    if (rank[px] == rank[py]) rank[px]++;
    return true;
}

// 4. DIJKSTRA - shortest path (non-negative weights)
int[] dist = new int[n];
Arrays.fill(dist, Integer.MAX_VALUE);
dist[start] = 0;
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]);
pq.offer(new int[]{start, 0});
while (!pq.isEmpty()) {
    int[] current = pq.poll();
    int node = current[0], w = current[1];
    if (w > dist[node]) continue;
    for (int[] nb : graph.get(node)) {
        int next = nb[0], w = nb[1];
        if (dist[node] + w < dist[next]) {
            dist[next] = dist[node] + w;
            pq.offer(new int[]{next, dist[next]});
        }
    }
}

// 5. BIPARTITE CHECK - BFS 2-coloring
int[] color = new int[n];
Arrays.fill(color, -1);
Queue<Integer> queue = new ArrayDeque<>();
for (int start = 0; start < n; start++) {
    if (color[start] != -1) continue;
    color[start] = 0;
    queue.offer(start);
    while (!queue.isEmpty()) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (color[neighbor] == -1) { color[neighbor] = 1 - color[node];
                queue.offer(neighbor);
            }
        }
    }
}

// 6. GREEDY
return true;
}
```

Greedy

Two Interval Templates

MERGE SELECTION RULE (for merge-type problems like #56, where both keys work):
Sort by START → traverse FORWARD (i = 0 → n-1), extend the END of last kept
Sort by END → traverse BACKWARD (i = n-1 → 0), extend the START of last kept
These two are exact mirror images. See #56 for both written out side by side.
(NOTE: this mirror only applies to merging. Template 1 below sorts by end but still sweeps FORWARD – there the goal is counting, not merging.)

TEMPLATE 1 – Sort by END time, sweep forward, track end of last kept interval
Use when: maximizing count of non-overlapping intervals
Greedy insight: earliest-finishing interval leaves most room for future ones

TEMPLATE 2 – Sort by START time, sweep forward, merge when overlapping
Use when: merging/counting overlapping intervals
Greedy insight: processing in start order → each interval only needs to compare with the last merged result

Variables:
intervals = list of [start, end] · lastEnd = end of last kept interval · result = merged output · pq = min-heap of active end times

Pseudocode:

TEMPLATE 1 (sort by end, count non-overlapping):
sort intervals by END time
lastEnd = -infinity
for each interval:
 if interval.start >= lastEnd: # no overlap
 lastEnd = interval.end
 else:
 skip it (overlap)

TEMPLATE 2 (sort by start, merge):
sort intervals by START time
for each interval:
 if result empty OR last kept end < interval.start:
 add interval as new
 else:
 extend last kept end = max(last end, interval.end)

TEMPLATE 3 (sort by start + min-heap of ends):
sort intervals by START time
for each interval:
 if heap nonempty AND earliest end <= interval.start:
 pop heap # reuse freed resource
 push interval.end # occupy a resource
heap size = resources needed

```
Arrays.sort(intervals, (a, b) -> a[1] - b[1]);
int lastEnd = Integer.MIN_VALUE;
for (int[] interval : intervals) {
    if (interval[0] >= lastEnd) {
        lastEnd = interval[1];
    }
}

Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
List<int[]> result = new ArrayList<>();
for (int[] interval : intervals) {
    if (result.isEmpty() || result.get(result.size()-1)[1] < interval[0]) {
        result.add(interval);
    }
    else {
        result.get(result.size()-1)[1] =
            Math.max(result.get(result.size()-1)[1], interval[1]);
    }
}

Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
PriorityQueue<Integer> pq = new PriorityQueue<>();
for (int[] interval : intervals) {
    if ((pq.isEmpty() && pq.peek() <= interval[0])
        || pq.poll());
    pq.offer(interval[1]);
}
```

Dynamic Programming

All Six Templates

Variables:
dp[i] = the answer for the subproblem at index i state i (meaning varies per family: ways/costs/length/feasibility ending at or reaching i) · n = problem size · target[] = knapsack capacity dimension.

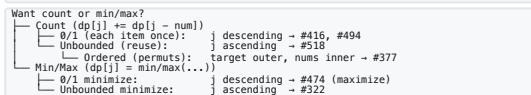
Pseudocode:

LINEAR 1D: dp[i] = fixed recipe over dp[i-1], dp[i-2] (Kadane: best ending at i = max(start fresh, extend))
LOOK-BACK: for each i, scan all earlier j and extend the best valid dp[j]
KNAPSACK: collapse item dimension to j; descending j = each item once, ascending j = reusable
2D SEQUENCE: match consumes both strings (diagonal), mismatch drops one side
INTERVAL: solve short ranges first, combine via a split point inside the range
GRID: each cell accumulates from cells it could arrive from (up / left)
STATE MACHINE: name a few states, update each from yesterday's states every step

```
// 1. LINEAR 1D - each cell depends on a fixed number of previous cells
int[] dp = new int[n+1];
dp[0] = base;
for (int i = 1; i <= n; i++)
    dp[i] = f(dp[i-1], dp[i-2], ...);

// 2. KADANE (max subarray, #53) = Linear-1D where dp[i] = best sum ENDING at i
// 2A. LOOK-BACK 1D - dp[i] depends on all j < i
int[] dp = new int[n];
for (int i = 0; i < n; i++) {
    for (int j = 0; j < i; j++) {
        dp[i] = f(dp[i], dp[j]);
    }
}

// 3. 2D SEQUENCE - match consumes both, mismatch drops one side. WHEN: LCS / edit
// 4. INTERVAL DP - short ranges first (i right-to-left, j from i+1; dp[i+1][w] ready).
// 5. GRID DP - each cell accumulates from cells it could arrive from (up/left).
// 6. STATE MACHINE - named states updated from yesterday's states each step.
```



Variables:
dp[i] = number of ways to reach sum i (count flavor; dp[0][1] is the empty selection); i = item index, j = current capacity/target, num = an item's weight/value. Loop direction decides reuse: descending j = item used at most once, ascending j = item reusable; target-outer = orderings counted.

Pseudocode:

0/1 KNAPSACK (each item once):
dp[0] = 1
for each item i:
 for j from target DOWN to nums[i]: // DESCENDING reads OLD dp (item i not yet used)
 dp[j] += dp[j - nums[i]]

UNBOUNDED KNAPSACK (item reusable):
dp[0] = 1
for each item i:
 for j from nums[i] UP TO target: // ASCENDING reads NEW dp (item i already used this pass)
 dp[j] += dp[j - nums[i]]

PERMUTATION COUNT (order matters):
dp[0] = 1
for j from 1 to target: // OUTER = target value
 for each num in nums: // INNER = try every num, so orderings differ
 if j >= num: dp[j] += dp[j - num]

// — 0/1 KNAPSACK — each item at most once
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++)
 for (int j = target; j >= nums[i]; j--)
 dp[j] += dp[j - nums[i]]

// — UNBOUNDED KNAPSACK — each item reusable
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++)
 for (int j = nums[i]; j <= target; j++)
 dp[j] += dp[j - nums[i]]

// — PERMUTATION COUNT — order matters (Combination Sum IV)
int[] dp = new int[target + 1];
dp[0] = 1;
for (int j = 1; j <= target; j++) {
 for (int num : nums) {
 if (j >= num) dp[j] += dp[j - num];
 }
}

Variables:
dp[i][j] = the answer for the prefixes s1[0..i] (first i chars) and s2[0..j] (first j chars); index 0 = empty prefix.

```
#121 (1 tx):          buy = max(hold, -price)          sell = max(hold, -price)          re-buy condition
#122 (unlimited):     hold = max(hold, cash - price)  hold = max(hold, cash - price)  no re-buy (ignore cash)
#123 (2 tx):         chain: buy2 uses s2[1] cash    chain: buy2 uses s2[1] cash    4 states chained
#309 (cooldown):     hold = max(hold, cooldown - price)  must wait 1 day
#714 (fee):          hold = max(hold, cash - price)      pay fee on sell
```

Pseudocode:

```
dp = (m+1) x (n+1) table
initialize dp[0][*] and dp[*][0] from the empty-prefix base cases
for i from 1 to m:
    for j from 1 to n:
        // each char of s1
        // each char of s2
        if s1[i-1] == s2[j-1]:
            dp[i][j] = dp[i-1][j-1] + 1 // current chars match
        else:
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]) // drop a char
shrink by 1;
return dp[m][n]
```

```
int[][] dp = new int[m+1][n+1];
for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {
        if (s1.charAt(i-1) == s2.charAt(j-1)) {
            dp[i][j] = dp[i-1][j-1] + 1;
        }
        else {
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]);
        }
    }
}
```

Variables:
dp[i][j] = the answer for the interval [i, j] (left endpoint i, right endpoint j).

Pseudocode:

```
dp = n x n table
for each i: dp[i][i] = base_case // length-1 intervals
for i from 1 to n-1 down to 0: // 1 RIGHT-TO-LEFT so dp[i+1][*] (inner) is ready
    for j from i+1 to n-1: // j LEFT-TO-RIGHT from i+1 so dp[i][j-1] is ready
        // safe to read dp[i+1][j], dp[i][j-1], dp[i-1][j-1] (all strictly smaller intervals)
        dp[i][j] = ...transition over interval [i,j]...
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;
for (int i = n-1; i >= 0; i--) {
    for (int j = i+1; j < n; j++) {
        dp[i][j] = ...;
    }
}
```

Variables:
dp[i][j] = the answer for the interval [j, i] (right endpoint i, left endpoint j).

Pseudocode:

```
dp = n x n table
for each i: dp[i][i] = base_case // length-1 intervals
for i from 0 to n-1: // 1 i RIGHT endpoint, FORWARD left-to-right
    for j from i-1 down to 0: // j = LEFT endpoint, DESCENDING from i-1 to 0
        // safe to read dp[i-1][j], dp[i][j+1], dp[i-1][j+1] (all shorter intervals already filled)
        dp[i][j] = ...transition over interval [j,i]...
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;
for (int i = 0; i < n; i++) {
    for (int j = i-1; j >= 0; j--) {
        dp[i][j] = ...;
    }
}
```

Variables:
dp[i][j] = the count/answer for the (i, j) subproblem, defined only for j <= i (lower-triangular).

Pseudocode:

```
dp = n x n table
for each i: dp[i][0] = 1 // column base case; dp[0][j=0] = 0
for i from 1 to n-1: // fill ROW BY ROW, top to bottom
    for j from 1 to i: // j from 1 up to i (triangular, j <= i)
        // safe to read dp[i-1][j], dp[i][j-1], dp[i-1][j-1] (prior row / earlier in row)
        dp[i][j] = ...transition...
```

```
int[][] dp = new int[rows][cols];
for (int i = 0; i < rows; i++) dp[i][0] = 1;
for (int i = 1; i < rows; i++) {
    for (int j = 1; j <= i; j++) {
        dp[i][j] = ...;
    }
}
```

Variables:
dp[i][j] = best (path count / cost) to reach cell (i, j) moving only right/down; memo[i][j] = longest path starting from (i, j) for the all-direction DFS variant; dr/dc = 4-direction deltas.

Pseudocode:

```
dr = {1,-1,0,0}; dc = {0,0,1,-1} // DOWN UP RIGHT LEFT
// Standard grid (only right/down - DAG, no cycle):
dp = rows x cols table
Initialize first row and first column
for i from 0 to rows-1:
    for j from 0 to cols-1:
        dp[i][j] = grid[i][j] + f(dp[i-1][j], dp[i][j-1]) // combine top & left
// All-direction grid (memoized DFS for increasing/decreasing paths):
memo = rows x cols table
for i from 0 to rows-1:
    for j from 0 to cols-1:
        dfs(grid, i, j, memo, dr, dc) // each cell caches its longest path
```

```
int[] dr = {1,-1,0,0}; dc = {0,0,1,-1};
int[][] dp = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++) {
        dp[i][j] = grid[i][j] + f(dp[i-1][j], dp[i][j-1]);
    }
int[][] memo = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++) {
        dfs(grid, i, j, memo, dr, dc);
    }
```

cash = max profit when NOT holding (free to buy or idle)
hold = max profit when HOLDING stock (bought it at some cost)
cooldown = max profit right after selling (can't buy today)

#121 (1 tx):	buy = max(hold, -price)	sell = max(hold, -price)	re-buy condition
#122 (unlimited):	hold = max(hold, cash - price)	hold = max(hold, cash - price)	no re-buy (ignore cash)
#123 (2 tx):	chain: buy2 uses s2[1] cash	chain: buy2 uses s2[1] cash	4 states chained
#309 (cooldown):	hold = max(hold, cooldown - price)	hold = max(hold, cooldown - price)	must wait 1 day
#714 (fee):	hold = max(hold, cash - price)	hold = max(hold, cash - price)	pay fee on sell

Bit Manipulation

Canonical Template

Variables:
result = XOR accumulator (lone survivor) · n = number being inspected/edited · i = bit position · count = set-bit counter · mask = 26-bit letter-preference set · masks: 1 << i selects bit i, n & (n-1) clears lowest set bit, n & (-n) isolates lowest set bit

Pseudocode:

```
XOR CANCEL:
result = 0
for each num in nums: result = result XOR num (pairs cancel, lone survives)
```

BIT OPERATIONS on n at position i:
isSet = bit i of n is 1 → (n >> i) & 1 == 1
set = turn bit i on → n | (1 << i)
clear = turn bit i off → n & ~(1 << i)
toggle = flip bit i → n ^ (1 << i)
lowest = isolate lowest set bit → n & (-n)
isPow2 = n > 0 and clearing lowest set bit gives 0

COUNT SET BITS (Brian Kernighan):
count = 0
while n != 0: count = count + 1; n = n & (n-1) (each step removes one set bit)

BITMASK AS SET:
mask = 0
for each char c in word: mask = mask | (1 << (c - 'a'))
two words share no letter when (mask[i] & mask[j]) == 0

```
// — XOR CANCEL — pairs cancel; lone element survives
int result = 0;
for (int num : nums) result ^= num;

// — BIT OPERATIONS —
boolean isSet = ((n >> i) & 1) == 1;
int set = n | (1 << i);
int clear = n & ~(1 << i);
int toggle = n ^ (1 << i);
int lowest = n & (-n);
boolean isPow2 = n > 0 && (n & (n-1)) == 0;

// — COUNT SET BITS — Brian Kernighan: each iteration removes one set bit
int count = 0;
while (n != 0) {
    count++; n = n & (n-1);
}

// — BITMASK AS SET — 26-bit integer represents presence of 26 letters
int mask = 0;
for (char c : word.toCharArray()) mask |= (1 << (c - 'a'));
if ((mask[i] & mask[j]) == 0) { /* no shared letter */ }
```

Variables:
ones/twos = bits seen 1/2 times mod 3 · xor = a ^ b of the two uniques · diff = lowest differing bit (xor & -xor) · a & b of the two unique values · dp[i] = set-bit counts by number · shift = bits dropped to reach common prefix · carry = AND-derived carry bits

Pseudocode:

VARIATION 1 (mod-3 state machine, Single Number II):
ones = 0; twos = 0
for each num: ones = (ones XOR num) AND NOT twos; twos = (twos XOR num) AND NOT ones
after loop, ones holds the element appearing once

VARIATION 2 (split XOR into two groups, Single Number III):
xor = XOR of all nums (= a XOR b)
diff = xor AND (~xor) (lowest bit where a and b differ)
a = 0; for each num: if (num AND diff) != 0: a = a XOR num (XOR one group)
b = xor XOR a (the other unique)

VARIATION 3 (DP relation, Counting Bits):
dp = array of size n+1
for i = 1..n: dp[i] = dp[i-1] + (i & 1 AND 1) (drop last bit, add it back)

VARIATION 4 (common prefix, Bitwise AND of Range):
shift = 0
while left != right: left >>= 1; right >>= 1; shift = shift + 1
return left << shift (shared high-bit prefix restored)

VARIATION 5 (carry loop, Sum of Two Integers):
while b != 0:
 carry = a XOR b (positions that carry)
 a = a XOR b (sum without carry)
 b = carry << 1 (carry moves one bit left)
return a

```
int ones = 0, twos = 0;
for (int num : nums) {
    ones = (ones ^ num) & ~twos;
    twos = (twos ^ num) & ~ones;
}

int xor = 0;
for (int num : nums) xor ^= num;
int diff = xor & (~xor);
int a = 0;
for (int num : nums)
    if ((num & diff) != 0) a ^= num;
int b = xor ^ a;
int[] dp = new int[n+1];
for (int i = 1; i <= n; i++)
    dp[i] = dp[i >> 1] + (i & 1);
int shift = 0;
while (left != right) {
    left >>= 1; right >>= 1; shift++;
}
return left << shift;

int add(int a, int b) {
    while (b != 0) {
        int carry = a & b;
        a = a ^ b;
        b = carry << 1;
    }
    return a;
}
```

Design

Canonical Template — HashMap + Doubly Linked List (LRU)

Variables:
map = key → CacheNode · head/tail = sentinels (head size = MRU, tail size = LRU) · node.previous/node.next = doubly linked neighbors

Pseudocode:

CacheNode: key, value, previous, next
Init: head <=> tail (head.next = tail; tail.previous = head)

remove(node):
node.previous.next = node.next
node.next.previous = node.previous

InsertAfterHead(node):
mark as most-recently-used
link node between head and head.next

evict LRU:
lru = tail.previous # node just before tail
remove(lru); map.remove(lru.key)

```
class CacheNode {
    int key, value;
    CacheNode previous, next;
    CacheNode(int key, int value) { this.key = key; this.value = value; }
}
// — SENTINEL NODES — head (MRU side) ← ... ← tail (LRU side)
CacheNode head = new CacheNode(0, 0), tail = new CacheNode(0, 0);
head.next = tail; tail.previous = head;
// — REMOVE NODE —
void remove(CacheNode node) {
    node.previous.next = node.next;
    node.next.previous = node.previous;
}
// — INSERT AFTER HEAD (= mark as MRU) —
void insertAfterHead(CacheNode node) {
    node.next = head.next;
    node.previous = head;
    head.next.previous = node;
    head.next = node;
}
// — LRU EVICTION — tail.previous is LRU
CacheNode lru = tail.previous;
remove(lru);
map.remove(lru.key);
}
```

Variations

Variables:
keyToVal/keyToFreq = key lookups · freqToKeys = freq -> ordered keys at that freq · minFreq = smallest live frequency · (RandomizedSet) List = values array · map = val -> index

Pseudocode:

```
VARIATION 1 (LFU): maps key->value, key->freq, freq->LinkedHashSet<key>, plus minFreq
on access: freq++; move key from freqToKeys[old] to freqToKeys[new]
on evict: remove first key (oldest) from freqToKeys[minFreq]

VARIATION 2 (RandomizedSet): ArrayList of values + HashMap val->index
remove: swap target with last element, update map, drop last
getRandom: list[random index]
```

Math

Canonical Templates

Variables:
n = the number being processed · digit = last digit peeled · x = base in fast power · result = running product/accumulator · b = the base for conversion · composite[] = sieve marks

Pseudocode:

```
DIGIT EXTRACTION:
while n is not zero:
    digit = remainder of n divided by 10
    drop the last digit of n
    use digit

FAST POWER (x^n):
result starts at 1
while exponent n still has bits:
    if the lowest bit is set, fold the current x into result
    square x to reach the next power
    shift n right to consume that bit
return result

BASE CONVERSION (number -> digits):
while num is positive:
    append num mod b
    divide num by b
reverse the collected digits
BASE CONVERSION (digits -> number):
value starts at 0
for each digit: value = value * b + digit

SIEVE:
make a composite[] flag array up to n
for i from 2 while i*i < n:
    if i already marked composite, skip it
    mark every multiple of i starting at i*i as composite
```

```
// — DIGIT EXTRACTION LOOP — peel digits right-to-left
while (n != 0) {
    int digit = n % 10;
    n /= 10;
}

// — FAST POWER (exponentiation by squaring) —
double fastPow(double x, long n) {
    double result = 1;
    while (n > 0) {
        if ((n & 1) == 1) result *= x;
        x *= x;
        n >>= 1;
    }
    return result;
}

// — BASE CONVERSION — generic base b, 0-indexed
StringBuilder builder = new StringBuilder();
while (num > 0) {
    builder.append(num % b);
    num /= b;
}
builder.reverse();
int value = 0;
for (int digit : digits) value = value * b + digit;

// — SIEVE OF ERATOSTHENES — mark composites up to n
boolean[] composite = new boolean[n];
for (int i = 2; i * i <= n; i++) {
    if (composite[i]) continue;
    for (int j = i * i; j <= n; j += i)
        composite[j] = true;
}
```